Birzeit University
Faculty of Engineering and Technology
Department of Electrical and Computer Engineering
ENCS4130 - Computer Network Laboratory (Term 1251)

Experiment#11: Design and Simulation of a Practice Topology Using Packet Tracer

Instructions:

- This is an **open manual** practice topology.
- You can work in group of **two** students at max.
- Follow the provided steps carefully and **complete all sections**.
- All tasks must be performed and demonstrated in **Cisco Packet Tracer**.
- Save your Packet Tracer file using your Student ID (SID) as the filename, in the following format: **SID.pkt** (For example, if your SID is **1222083**, name the file: **1222083.pkt**).
- **Save your work regularly**—you are responsible for preventing data loss.

Section A	Topology Building and Subnetting

Build the topology shown in **Figure 1** while meeting the following requirements:

1) Use the following **devices** in your implementation:
   A) Routers (**Router-2811**). B) Switches (**Switch-PT**).
   C) Multilayer Switch (**3560-24PS**). D) Servers (**Server-PT**).
   E) Access Point (**AccessPoint-PT**).
   F) PCs/Laptops/Tablets/Printers (**PC/Laptop/TabletPC/Printer-PT**).
2) Use the "**Automatically Choose Connection Type**" to connect devices.
3) If needed, add interfaces (e.g., **NM-1FE-TX**) to the **Router-2811** devices.

   The IPv4 and IPv6 addresses for each network are shown in **Figure 1**. You need to use **192.Y.X.0/22** for IPv4, replace the **variable Y** in the network ID with two-digit values derived from your SID (e.g. **if your SID is 1212083, then Y = 20**) and the **variable X** is defined by your instructor in the lab.
4) Do the **IPv4** subnetting for the given topology based on following steps:
   a. Count required subnets.
   b. Check hosts per subnet.
   c. Choose the right subnet size: use VLSM and start with the subnet that needs the most hosts and pick the correct mask. Don't forget that you have network IP and broadcast IP.
   d. Create subnet ranges: write network address, usable IP range, and broadcast address. You need to add it in the packet tracer file as notes for each subnet.

   The following table summarizes the available subnets and the number of hosts per subnet:

Subnet#	Number of hosts
Network1	*50*
Network2	2
Network3	20

Network4	2
Network5	2
Network6	28
Network7	20
Network8	14

Note that the router interfaces and end devices are included in the host count for each network in the table.

Section B	Internet Protocol Version 4 (IPv4), IPv6, and DHCP Configuration

Configure the **IPv4 addresses** using the results from the subnetting part and the assigned **IPv6 addresses** in the given topology:

1) Configure **Router R2** as a **DHCP server** for the following networks:
   - **IPv4: from the subnetting part**
   - **IPv6: 2001:Y:X:33::/64**

   Excludes the first five IPv4 addresses in the pool for reserved purposes (e.g., the default gateway, DNS server, Web server, and future expansions).
2) Assign **dynamic IPv4 and IPv6** addresses to **Laptop1** using DHCP.
3) Assign appropriate **static IPv4 (and IPv6) addresses** to all other end devices (i.e., PCs, laptops, servers, printers, and tablets) and needed interfaces.

   Note: Subnets within the **AS 100** must be configured with both IPv4 and IPv6 addresses.
4) Ensure proper configuration of subnet masks, prefix lengths, default gateways, and DNS server settings for all end devices.
5) Configure the access point as follows:

Parameter	Value
Name	*Your First Name*
Security Model	WPA2-PSK
Pre-Shared Key (PSK)	enCS4130-*SID* *(Replace SID with your student ID)*
Encryption Type	AES

6) Connect **Tablet0** wirelessly through the access point.

Section C	Servers Configuration

1) Configure **Server0** as a **DNS server**.
2) Configure **Server1** as a **Web server** with the domain name **www.ramallah.com**. Set the DNS server to resolve this domain name to its corresponding IPv4 and IPv6 addresses.
3) Configure **Server2** as an **Email server** with the domain name **ramallah.com**. Set the DNS server to resolve this domain name to its corresponding IPv4 and IPv6 addresses. Create two user accounts as follows:

Account Number	User	Password
1	*Your First Name*	*Your SID*
2	*Your Last Name*	*Your SID*

4) Configure the email client on **Laptop0** to use the first email account, and the email client on **Laptop1** to use the second email account.

Section D	Switching and Creating Virtual Local Area Networks (VLANs)

Configure the VLANs as shown in **Figure 1**, ensuring the following requirements are met:

1) **Computer Science Department (VLAN 10):**
   - **IP Address:** from the subnetting part.
   - **Gateway:** Multilayer Switch **MLS0**.
2) **Electrical Engineering Department (VLAN 20):**
   - **IP Address:** from the subnetting part.
   - **Gateway:** Router **R0**.
3) **Computer Engineering Department (VLAN 30):**
   - **IP Address:** from the subnetting part.
   - **Gateway:** Router **R0**.

Perform the necessary configurations on the switches to ensure proper setup for the VLANs and their associated ports.

Section E	IPv4/IPv6 Routing Configuration

Configure the routing protocols as follows:

1) **OSPF for IPv4:**
   - Use open shortest path first (**OSPF**) for IPv4 routing within each autonomous system:
     - **AS 100:** Configure three areas (**Area 0**, **Area 1**, and **Area 2**).
     - **AS 200:** Configure a single area (**Area 0**).
2) **BGP for IPv4:**
   - Use border gateway protocol (**BGP**) for IPv4 routing between **AS 100** and **AS 200**.
3) Perform the necessary **redistribution** between **OSPF** and **BGP** to ensure seamless connectivity across the topology.
4) **Static Routing for IPv6:**
   - Configure **static routes** to handle IPv6 traffic within **AS 100**.

At the end of this section, **ensure that the entire topology is fully connected**.

Section F	Access Control List (ACL) Configuration

Use ACLs to implement the following restrictions efficiently:

1) Use a standard ACL to block **Laptop1** from accessing **VLAN 10**.
2) Use an ACL to block *HTTP traffic* originating from **VLAN 30**.

Ensure that the ACLs are **configured on the correct devices** and **applied to the appropriate interfaces in the correct direction** (inbound or outbound).

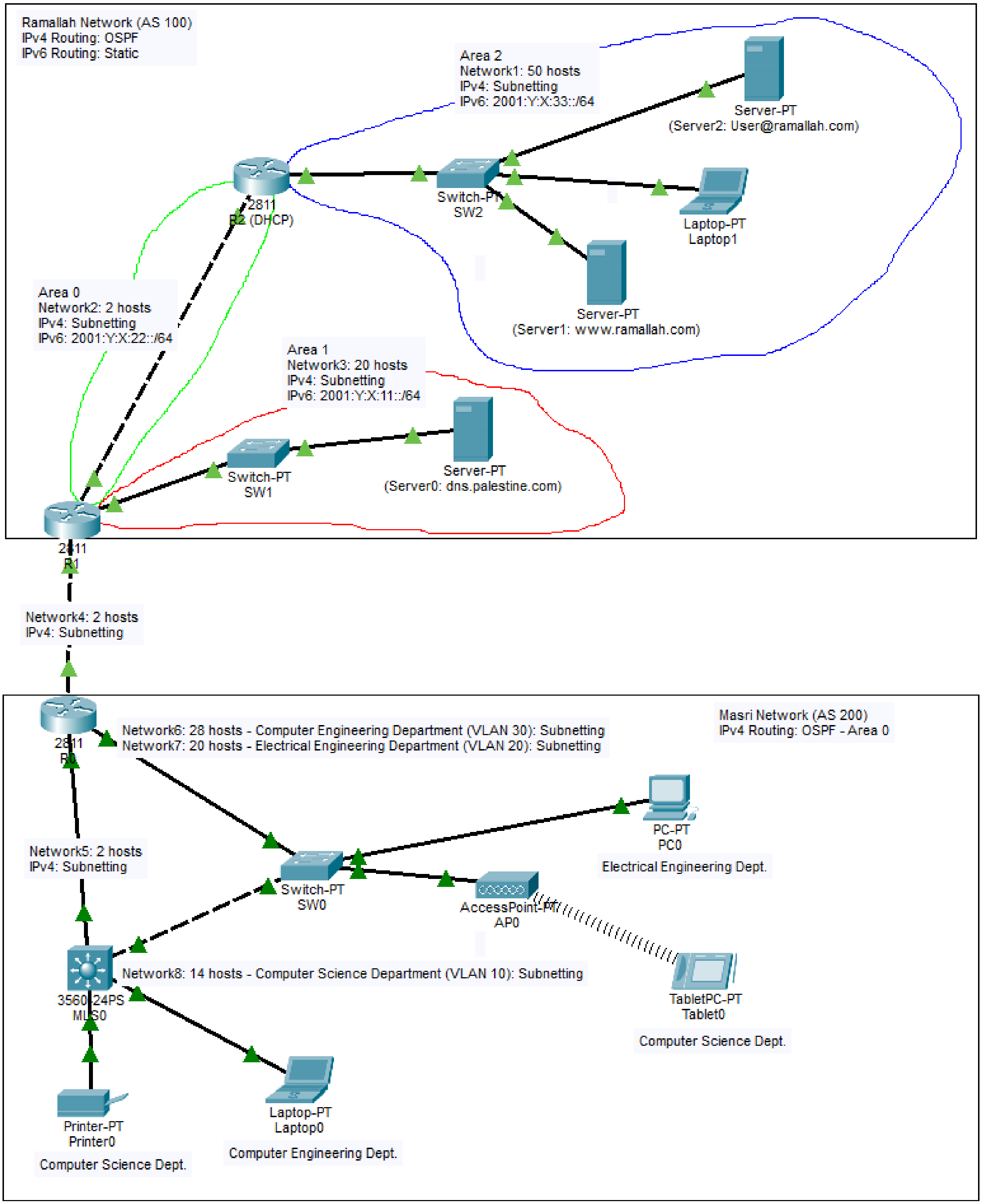

Figure 1: Network Topology.